



# Quantum Error Correction Decoders for 2D Color Codes

## Systematic Simulation and Comparison

ANTONIO MORALES GARCÍA

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### Abstract

This Master's Thesis presents a comprehensive study on quantum error correction (QEC) specifically focused on 2D color codes. Using state-of-the-art simulation tools like Stim and Py-matching, we analyze decoder performance on 4.8.8 lattice geometries. The study concludes with a comparative analysis of the results and explores potential improvements for the decoding process.